

Network data

Networks and Innovation Training Programme
Dr Antoine Vernet & Dr Anne ter Wal

September 2013

Learning objectives

At the end of this session, you will...

- ...be familiar with various ways through which **existing data** can be interpreted **as network data**
- ...understand existing and novel **approaches to collect network data** yourself
- ...be aware of the need to **carefully select** relevant **covariates** to complement network data

Interpreting data as networks

- Patent data
- Publication data
- Collaboration data (FP7)
- Project membership data
- Licensing data
- Labour mobility data
- Trade flows
- Data from social networks (e.g. Twitter)
- Community data
- Investment data

Patent data

INTERPRETING EXISTING DATA AS NETWORK DATA

(12) **United States Patent**
Dutta

(10) **Patent No.:** **US 6,789,076 B1**
(45) **Date of Patent:** **Sep. 7, 2004**

(54) **SYSTEM, METHOD AND PROGRAM FOR
AUGMENTING INFORMATION RETRIEVAL
IN A CLIENT/SERVER NETWORK USING
CLIENT-SIDE SEARCHING**

(75) Inventor: **Rabindranath Dutta**, Austin, TX (US)

(73) Assignee: **International Business Machines
Corp.**, Armonk, NY (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **09/569,503**

(22) Filed: **May 11, 2000**

(51) Int. Cl.⁷ **G06F 17/30**

(52) U.S. Cl. **707/5; 707/10**

(58) Field of Search **707/5, 10, 2, 3,
707/4, 200; 709/203, 200**

(56) **References Cited**

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6,480,835 B1 * 11/2002 Light 707/3

OTHER PUBLICATIONS

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file Information," Mar. 1988.

IBM Technical Disclosure Bulletin, "Searching for Docu-
ments that do not have a Standalone View," Oct. 1989.

IBM Technical Disclosure Bulletin, "Contextual Data for
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IBM Technical Disclosure Bulletin, "Information Retrieval
System Interface," Nov. 1989.

IBM Technical Disclosure Bulletin, "Subsetting of Direc-
tory Partitions and Conditional Use for Query," May 1991.

(List continued on next page.)

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(57) **ABSTRACT**

In a system, method and program for identifying informa-
tion accessible over a client/server network, search results
obtained by a search engine running on a server may be
improved by utilizing computing resources on a client.
Initial search results produced by the server search engine
are further processed on the client machine, which may
produce a more refined set of search results. The processing
on the client machine may be done using a client-side search
program. The usefulness of the search results to a user may

Patent data

INTERPRETING EXISTING DATA AS NETWORK DATA

Applicants

alpha

beta

gamma

delta

Patents

1

2

3

4

5

Inventors

A

B

C

D

E

F

G

H

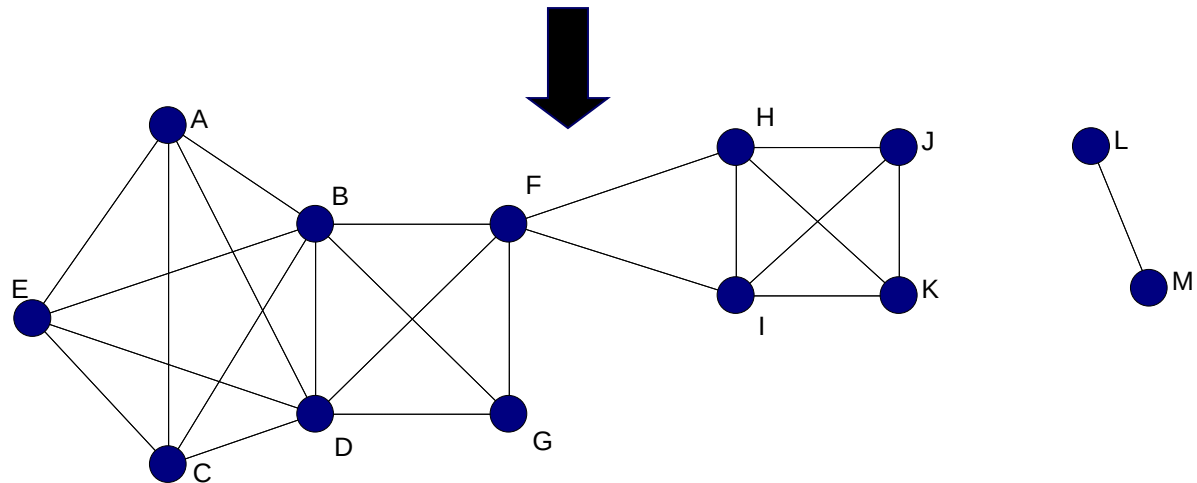
I

J

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L

M



Patent data

INTERPRETING EXISTING DATA AS NETWORK DATA

PROPERTY ORGANIZATION

Scheme RCL Compilation Catchwords

A SECTION A — HUMAN NECESSITIES

HEALTH; LIFE-SAVING; AMUSEMENT

A63 SPORTS; GAMES; AMUSEMENTS

A63B APPARATUS FOR PHYSICAL TRAINING, GYMNASTICS, SWIMMING, CLIMBING, OR FENCING; BALL GAMES; TRAINING EQUIPMENT (appa

Swimming

A63B 35/00 Swimming framework with driving mechanisms operated by the swimmer or by a motor (other vessels or like floating structures for pleasure or sport B63B 35/71, E

- A63B 35/02 • shaped like a fish tail
- A63B 35/04 • with paddle wheels
- A63B 35/06 • with twin-bladed paddles or buoyant members
- A63B 35/08 • with propeller propulsion

<http://www.wipo.int/classifications/ipc/en/>

Patent data

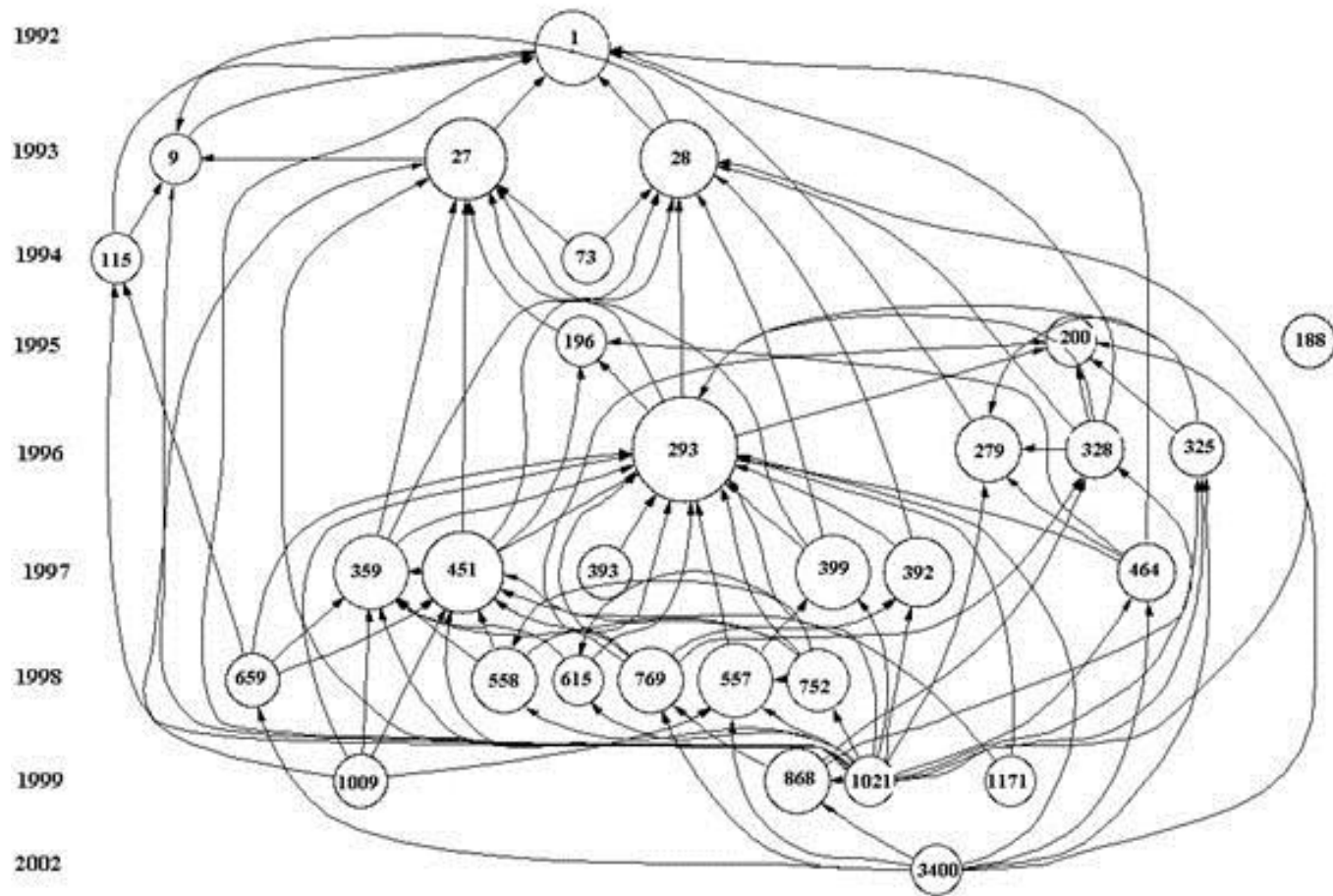
INTERPRETING EXISTING DATA AS NETWORK DATA

Measures of patent quality:

- **Granted** patents versus patent applications
- **Renewed patents**
 - only around 10% of the patents issued in Germany, France and the UK 'lives' for the full statutory term
- **Number of claims**: indication that an innovation is broader and of greater potential profitability
- **Patent family size**: determines the geographic coverage of the protection and can therefore be used as a proxy of the importance the patent owner places on their innovation
- **Citations received**: importance of the patent in the eyes' of others

Citation data – main path analysis

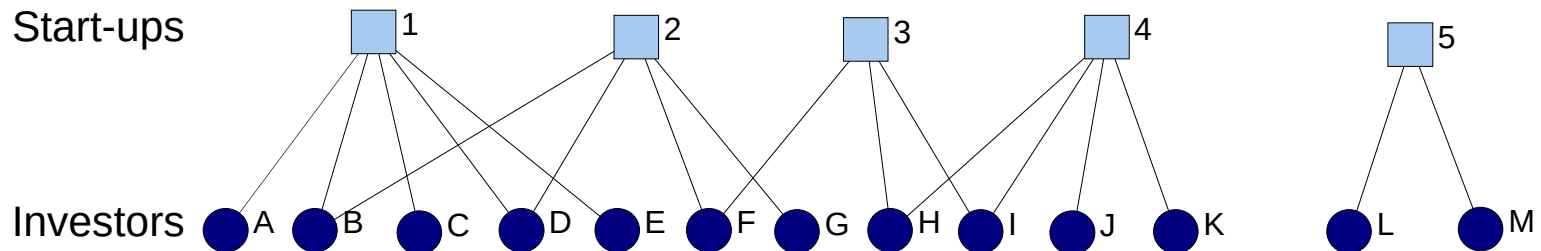
INTERPRETING EXISTING DATA AS NETWORK DATA



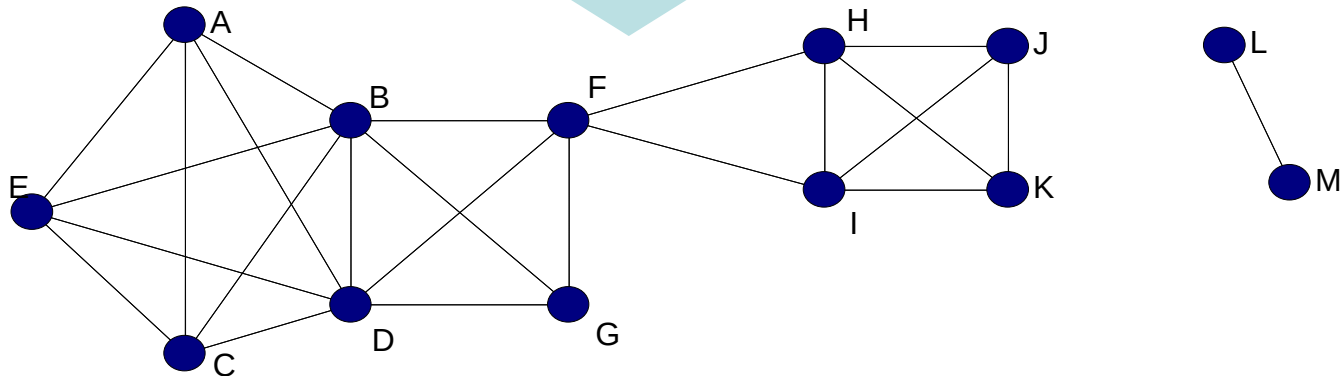
Source: Lucio-Arias & Leydesdoff (forthcoming)

Investment data

Investment relations of investors in start-ups (two-mode)



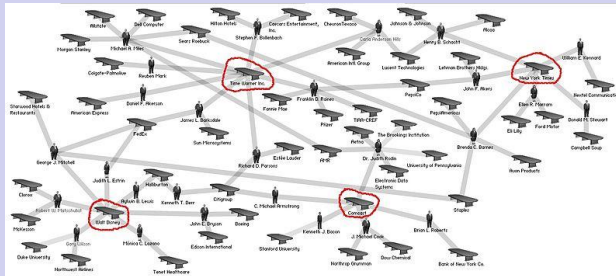
Syndication network among investors (one-mode)



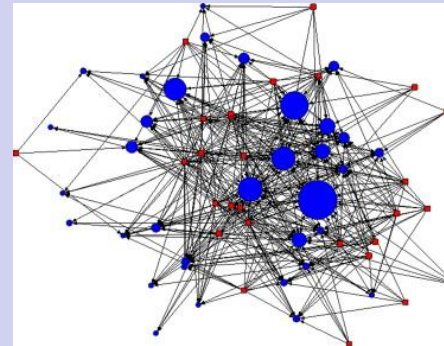
Other affiliation data

INTERPRETING EXISTING DATA AS NETWORK DATA

Interlocking directorships



Project memberships



Event attendance



Past shared employment



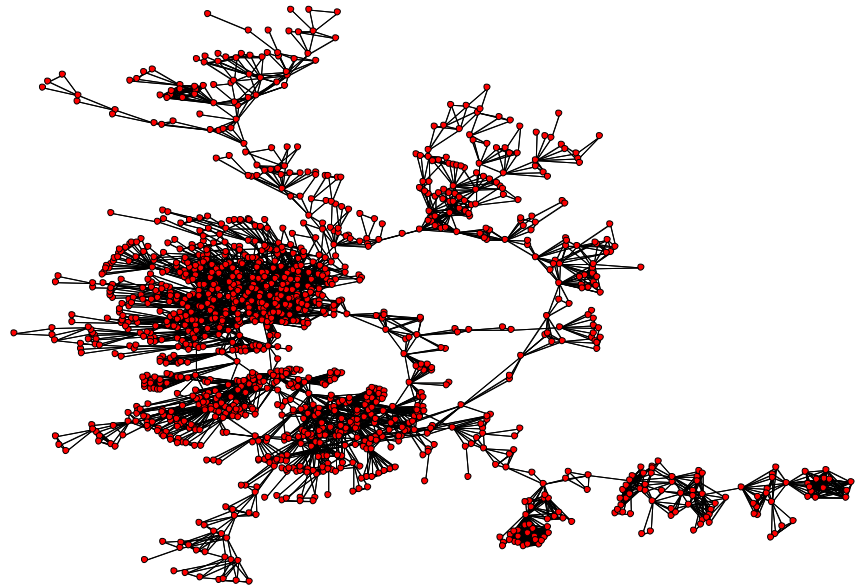
A note on two-mode data

INTERPRETING EXISTING DATA AS NETWORK DATA

**Even when projected as one-mode networks
the two-mode nature of affiliation data affects the
topology of networks**

Data structure affected by:

- Distribution of number of projects per actor
- Distribution of number of actors per project

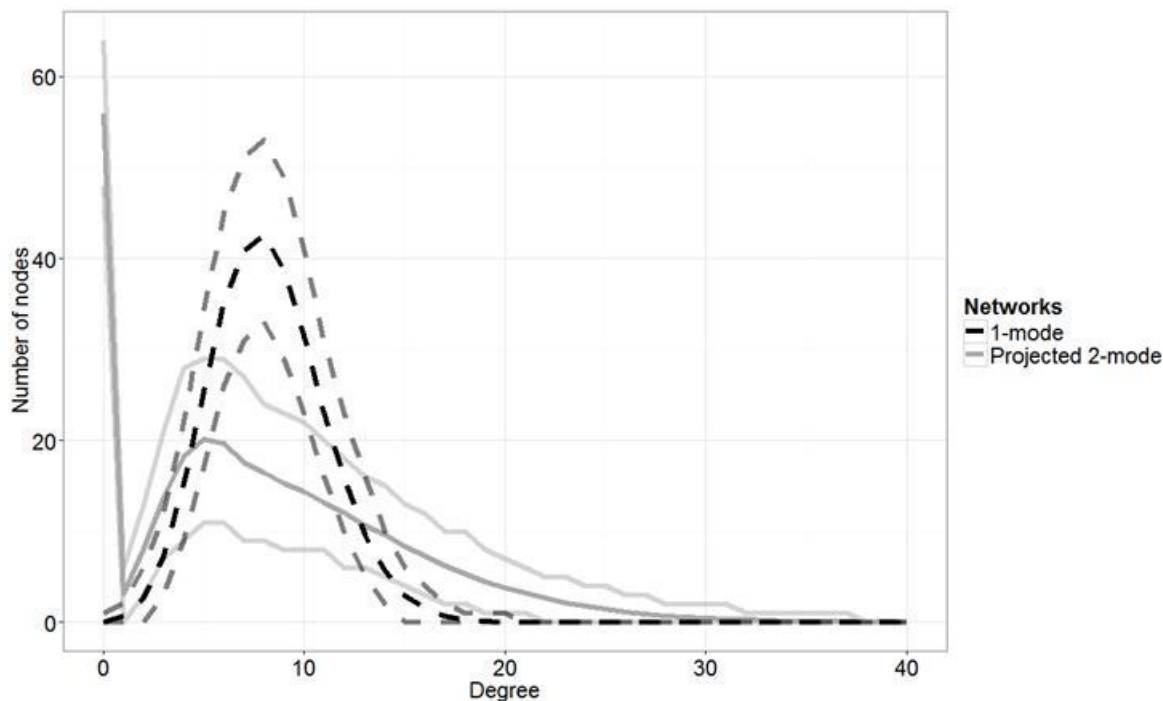


Source: *picture from own research*

Two mode networks

One can never be too cautious

- Two mode networks are intrinsically different from one mode networks



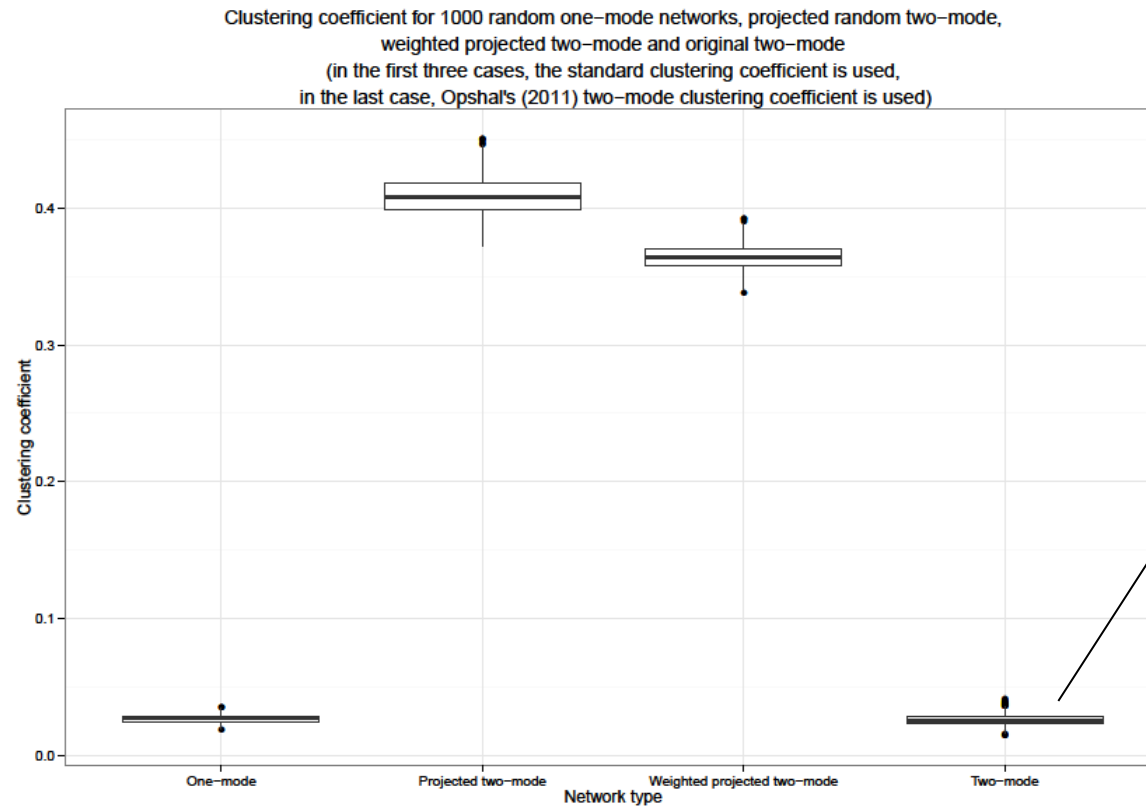
Different degree distributions

Projection make this worse

Two mode networks

One can never be too cautious

Clustering coefficient

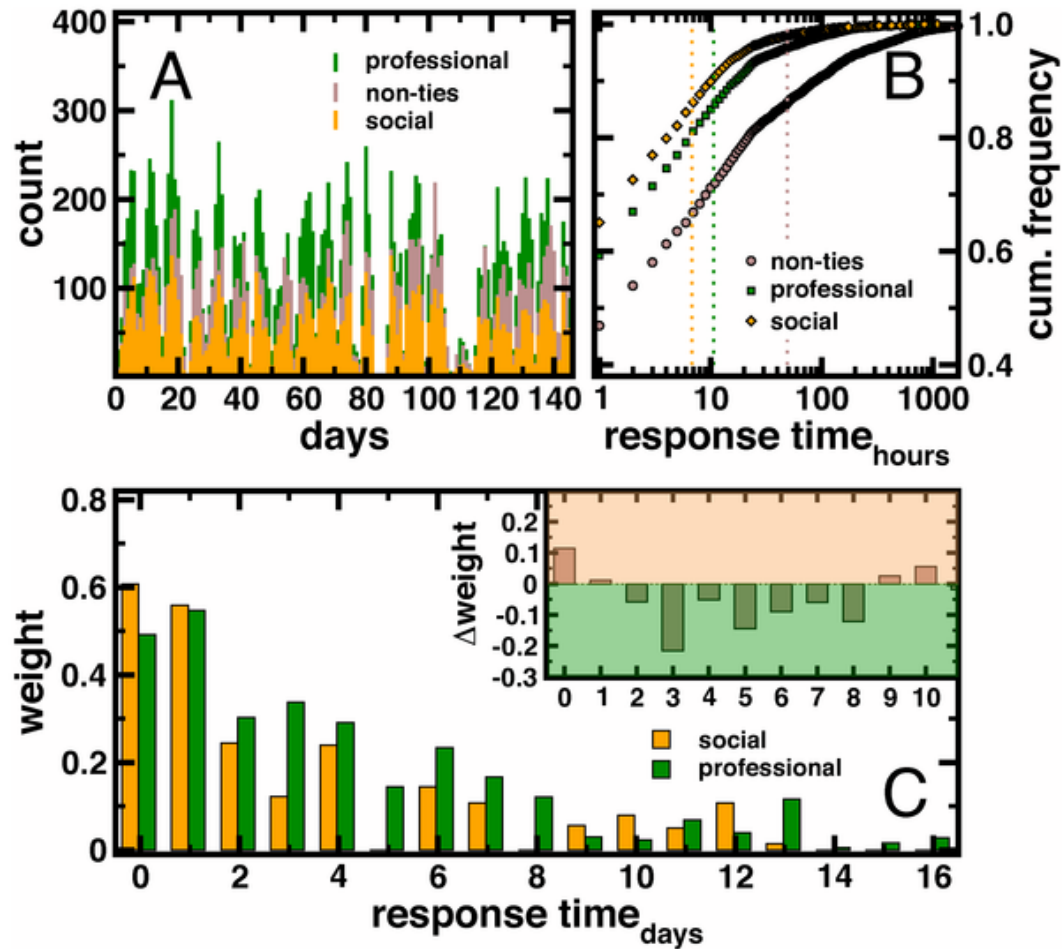


Projection create spurious clustering

Measures designed for two mode data can help solve this problem

Email data

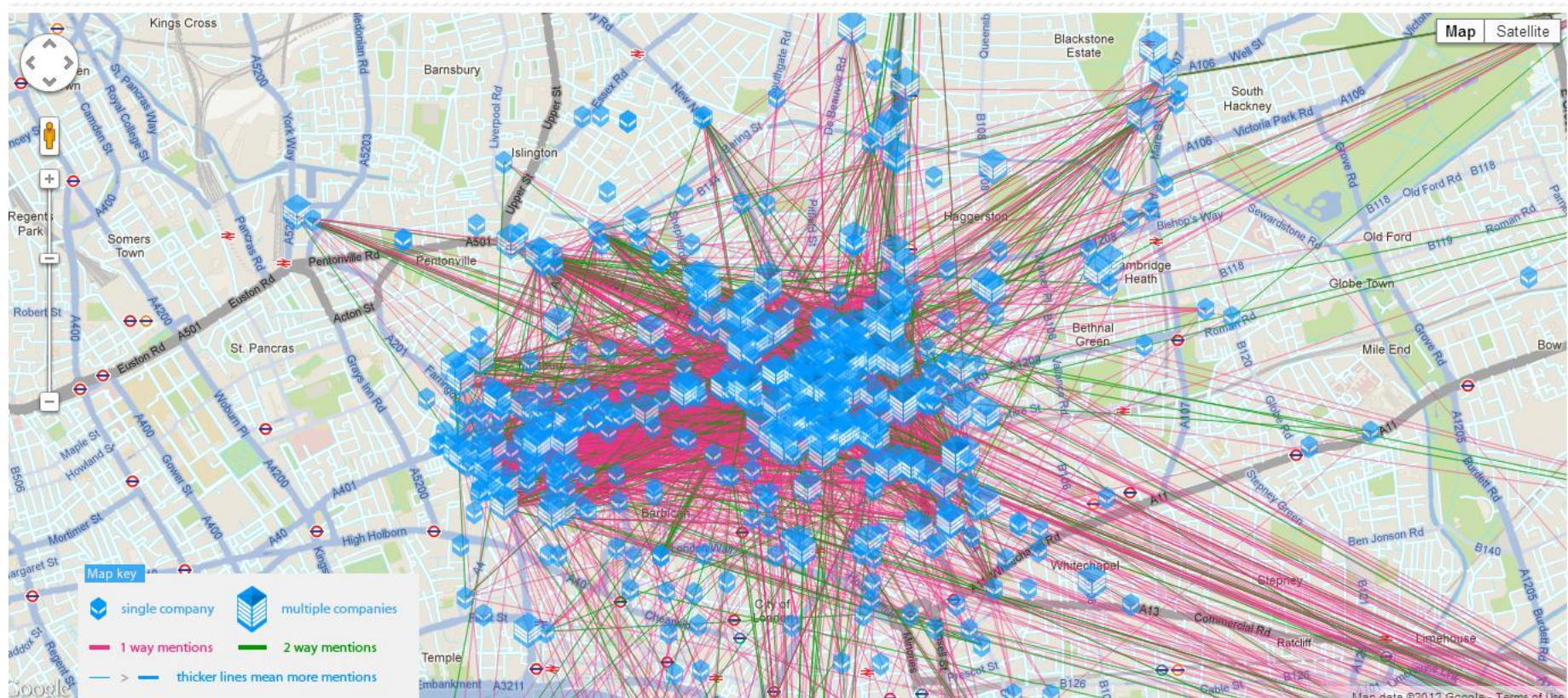
INTERPRETING EXISTING DATA AS NETWORK DATA



Source: Wuchty & Uzzi (2011, PLoS one)

Social network data

INTERPRETING EXISTING DATA AS NETWORK DATA



Online communities

INTERPRETING EXISTING DATA AS NETWORK DATA

- Forums
- Developers communities (SourceForge)
- Twitter
- Q & A sites



What is **sourceforge** ?

- A platform to:
 - provide tools to develop open source software (under three different GPLs)
 - bring together developers
 - distribute the software created
- A city of software developers
 - Over 260,000 projects
 - 2.5 million developers
 - 2 millions downloads per day
- Milestones



Online communities

INTERPRETING EXISTING DATA AS NETWORK DATA

- Network data on 
 - Affiliation (members of projects)
 - Communication (on the forums)
 - Technical interaction (bug tracking system)
- Network data on forums
 - Thread analysis (who attaches their own comments to the thread)
 - Message reply analysis (for example through explicit markers like: @)

Online communities

Interpreting Existing Data as Network Data

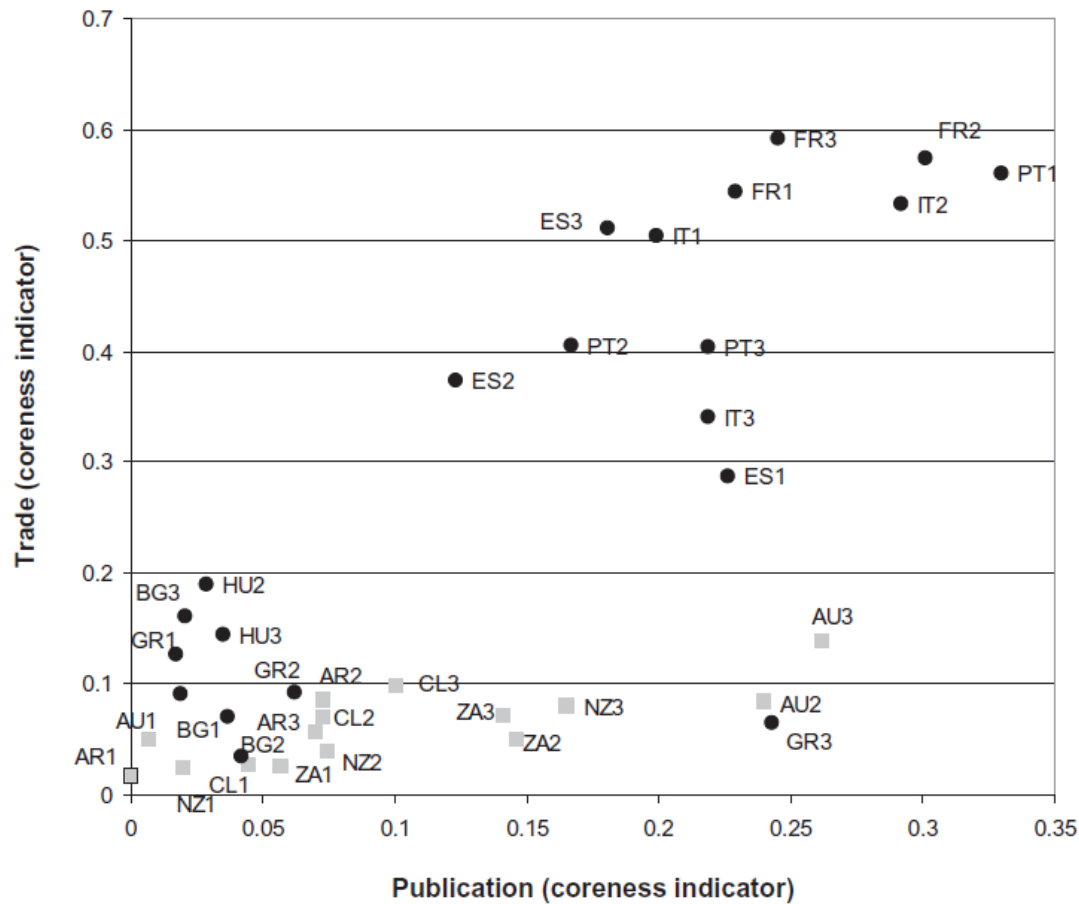
- A word of caution on the data structure in the previous cases:
 - Because the data is very rich, there are many ways of building networks from it
 - The data often exhibit a two mode structure

Multiplexity

Multilevel

Trade networks

INTERPRETING EXISTING DATA AS NETWORK DATA



Source: Cassi, Morrison & Ter Wal (2012)

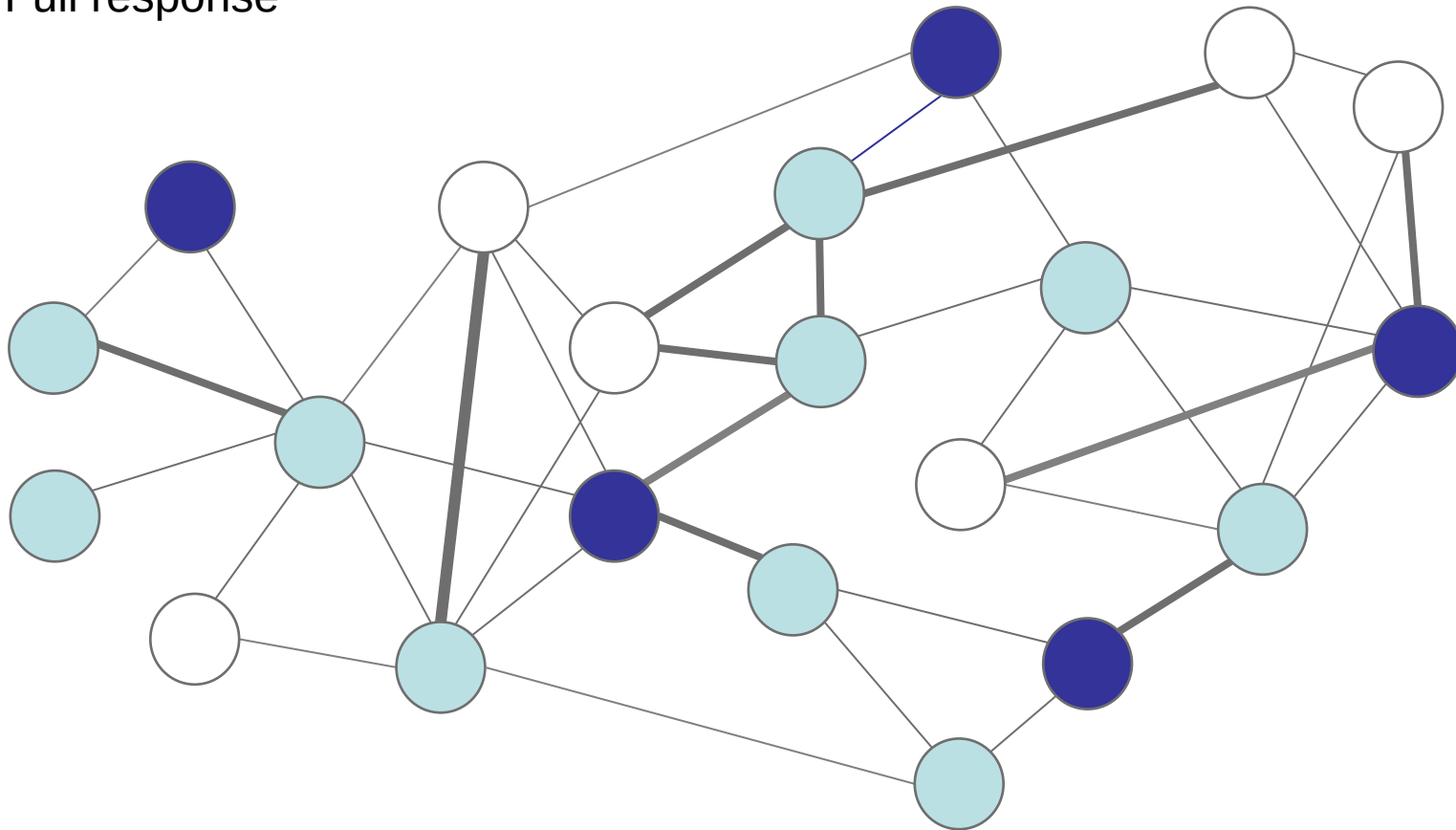
Network data

COLLECTING NETWORK DATA

Ego versus full network data

COLLECTING NETWORK DATA

Full response

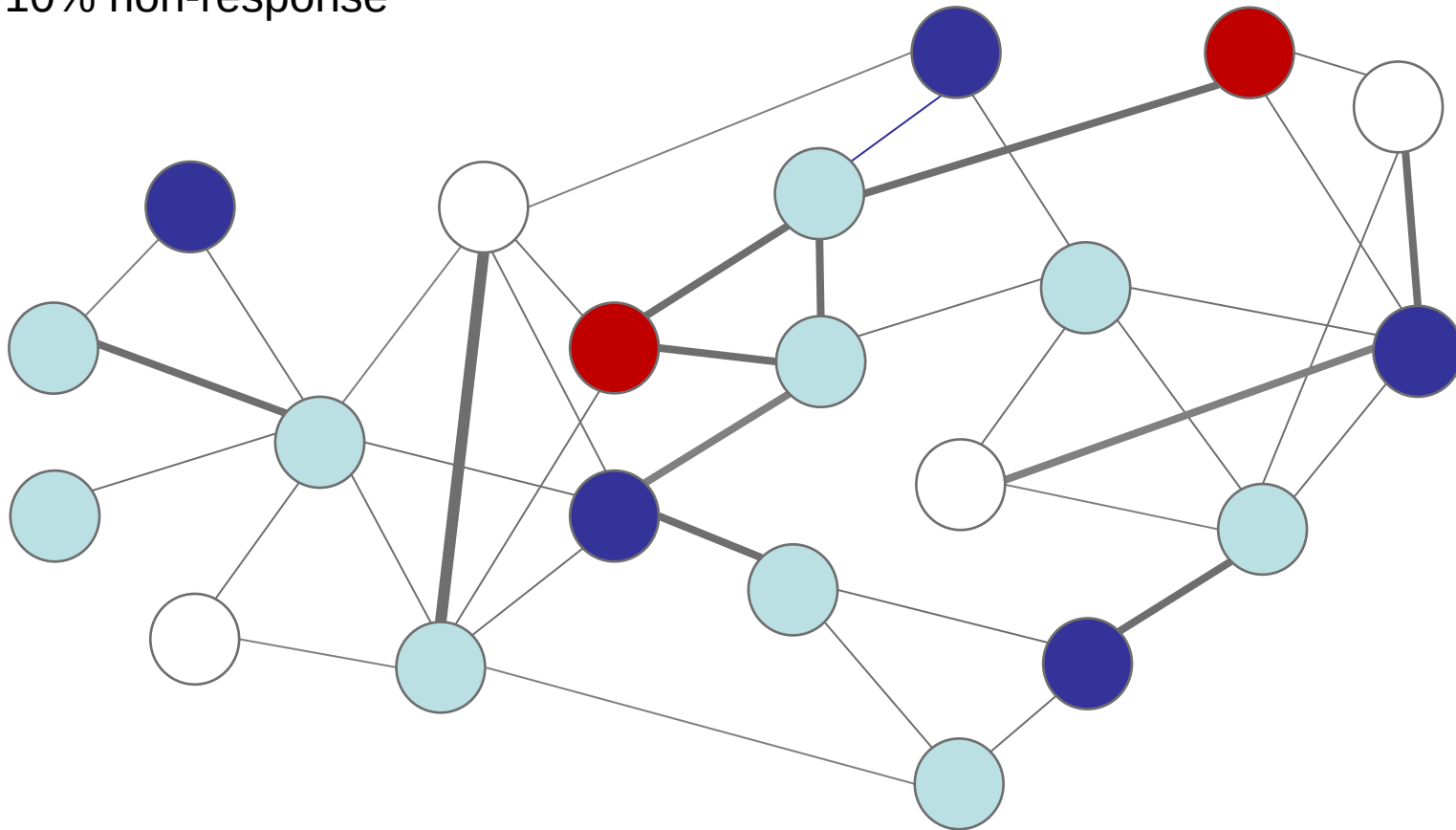


Source: Ter Wal & Boschma (2009)

Ego versus full network data

COLLECTING NETWORK DATA

10% non-response

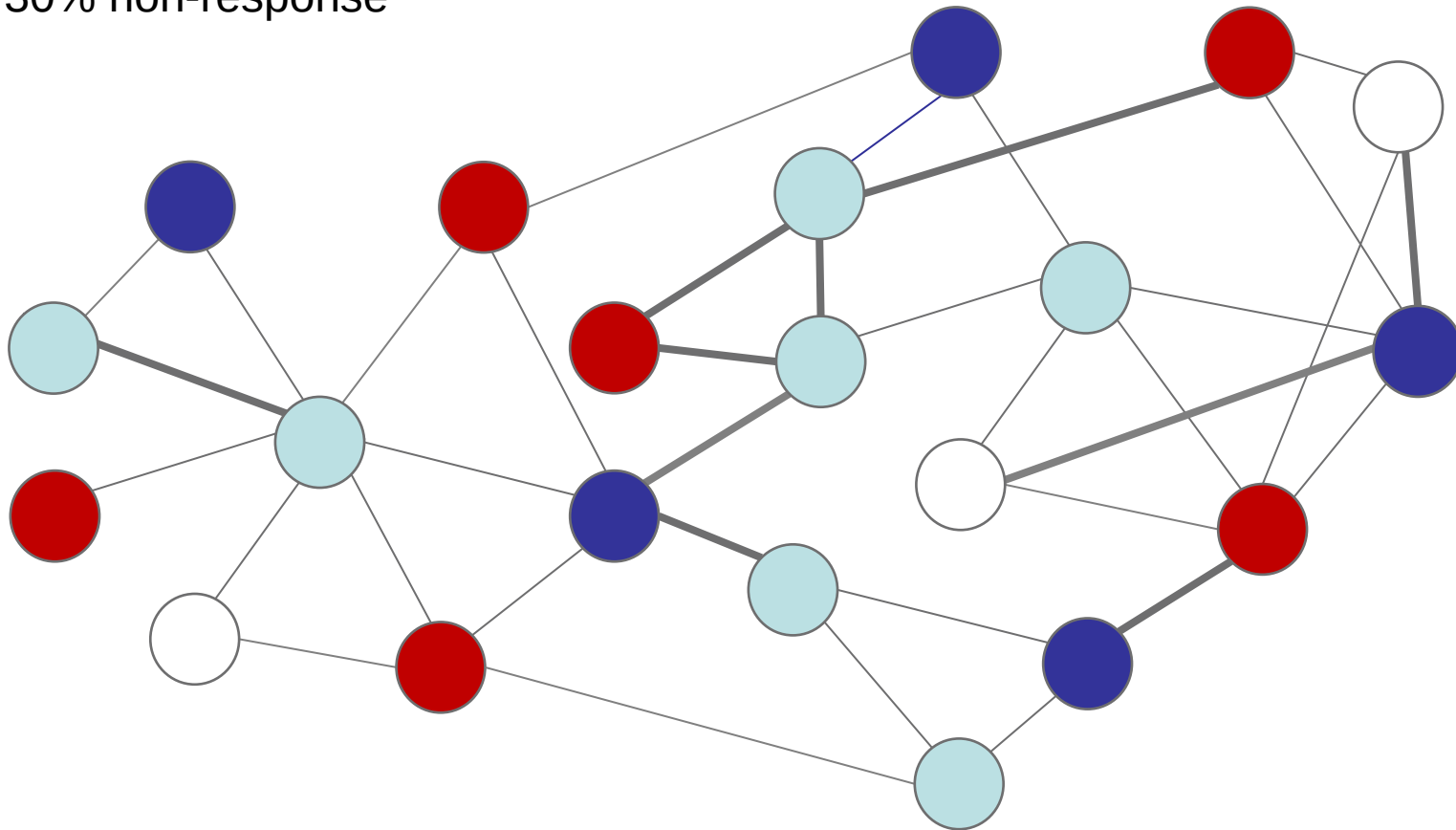


Source: Ter Wal & Boschma (2009)

Ego versus full network data

COLLECTING NETWORK DATA

30% non-response



Source: Ter Wal & Boschma (2009)

Ego versus full network data

COLLECTING NETWORK DATA

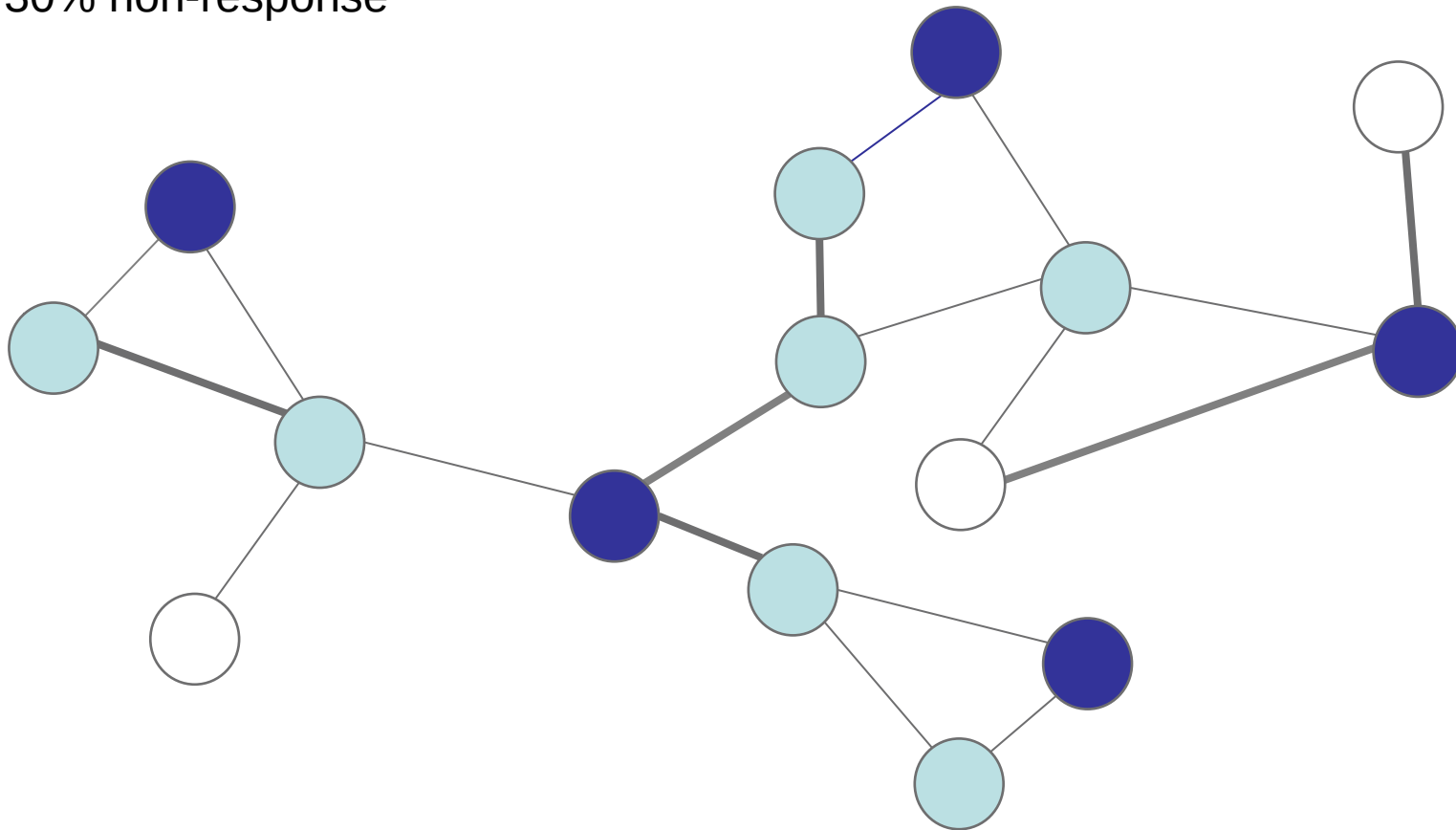
This is under the assumption that:

- Ties are undirected
- A tie is always reported by both parties involved.

Ego versus full network data

COLLECTING NETWORK DATA

30% non-response



Name generator

COLLECTING NETWORK DATA

8. Please specify the names of your individual contacts below by answering the following questions. The individuals you mention can be either from within or outside your organization. It is sufficient to give first name and first initial of last name (e.g. Frank M.).

Over the last six months, are there any work-related contacts from whom you regularly sought information and advice as input for your research and development work?

1.
2.
3.

Suppose you were moving to a new job and wanted to leave behind the best network advice that you could for the person moving into the current job. Are there any individuals whom you would name whose knowledge and expertise is critical as a source of new ideas?

1.
2.
3.

Is there any work-related contact who you regard as a source of support for implementing your new ideas – that is, someone you are comfortable with discussing your new ideas?

1.
2.
3.

Please list anyone who has been an important source of new knowledge and ideas to you over the last six months, who you have not listed in the previous questions.

1.
2.
3.

Name interpreter

COLLECTING NETWORK DATA

9. Please complete the table below by answering for each person:

- How frequently you are in touch (column 1)
- Where they are employed (column 2, 3 or 4)

	Frequency of contact	If same business unit	If other business unit	If outside your organization
Chris B.	▼	▼	▼	▼
	☐ Rarely ☐ A few times a year ☐ Monthly ☐ Weekly ☐ More than once a week	☐ Peer technologist ☐ Junior technologist ☐ Senior technologist ☐ Peer manager ☐ Junior manager ☐ Senior manager ☐ Other internal	☐ Business Unit 1 ☐ Business Unit 2 ☐ Business Unit 3 ☐ Business Unit 4 ☐ Business Unit 5 ☐ Business Unit 6	☐ Academic ☐ Consultant ☐ Customer ☐ Supplier ☐ Individual inventor ☐ Competitor ☐ Non-competing firm ☐ Private research inst. ☐ Innovation broker ☐ Professional org. ☐ Other

Alter-alter ties

COLLECTING NETWORK DATA

10. Check the box for any two people who know each other in person.

	Chris B	Paul B	Colin E	Melanie E	Colin H
Chris B		☐	☐	☐	☐
Paul B			☐	☐	☐
Colin E				☐	☐
Melanie E					☐

Measuring tie strength

COLLECTING NETWORK DATA

- **Frequency of contact**
 - 5-6 categories, ranging from “rarely” to “more than once a week”
 - Duration of a relationship measured as the natural logarithm of years of acquaintance
- **Emotional intensity / Closeness**
 - Status as acquaintance, good friend or close friend
- **Degree of intimacy**
 - Breadth of discussions, measured as whether or not discussing family, friends, politics, local events, work, and leisure
 - Mutual confiding, indicated by whether or not they confided in someone or someone confided in them about personal problems (or voting advice)
- **Reciprocal support services**
 - No measurement provided in Marsden and Campbell (1984). They refer to Wellman (1982) in this regard.

Perceived versus actual network

COLLECTING NETWORK DATA

Which one does theoretically matter?

- Is it the network resources people have access to?
- Or the network resources they think they have access to?

Biases in the perception of networks

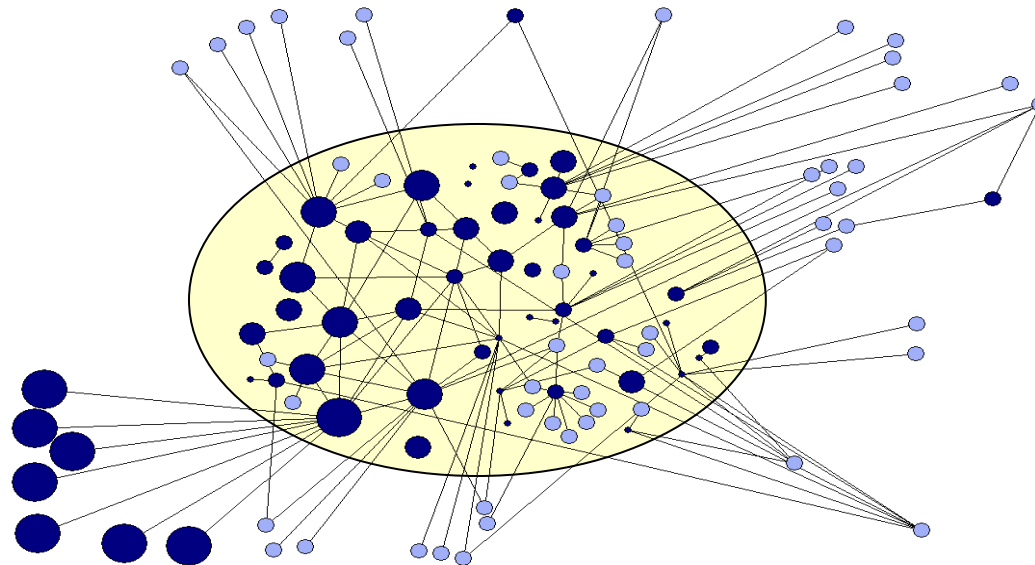
- Leaders may perceive themselves more central than they are.
- Popular actors may be perceived as more popular than they are.
- Individuals may perceive friendship networks as small worlds
- Second-order links (beyond ego's own social circle) is challenging

Perceived to have a prominent friend boosted individual reputation, whereas actually having a prominent friend had no effect.

The boundary-specification problem

COLLECTING NETWORK DATA

Specifying the boundaries of a network is not an empirical question, but a theoretical one.

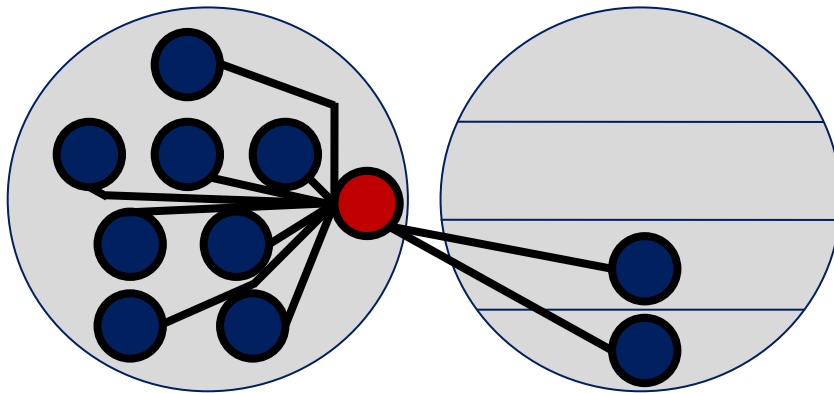


Source: Borgatti & Halgin (2011) – image from own research (Boschma & Ter Wal 2007)

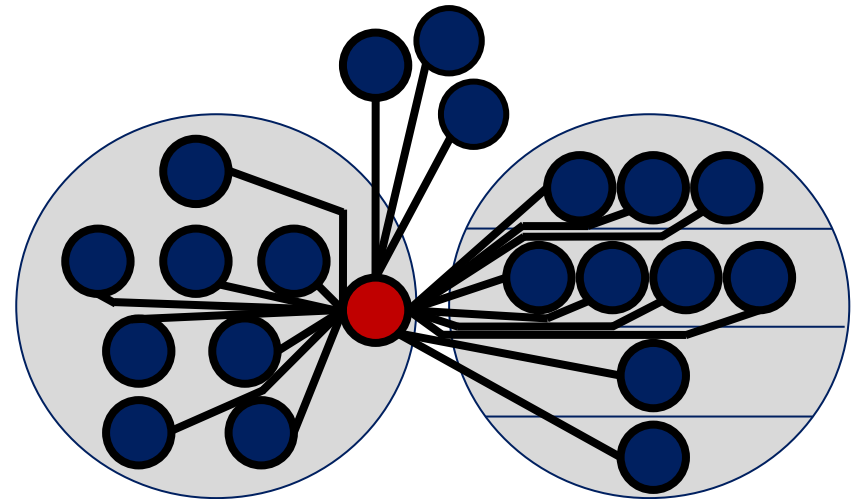
Biasing responses

COLLECTING NETWORK DATA

Measuring internal and external ties at once



Measuring internal and external ties separately



Network data

THE SEARCH FOR RELEVANT COVARIATES

Typology of covariates

COVARIATE DATA

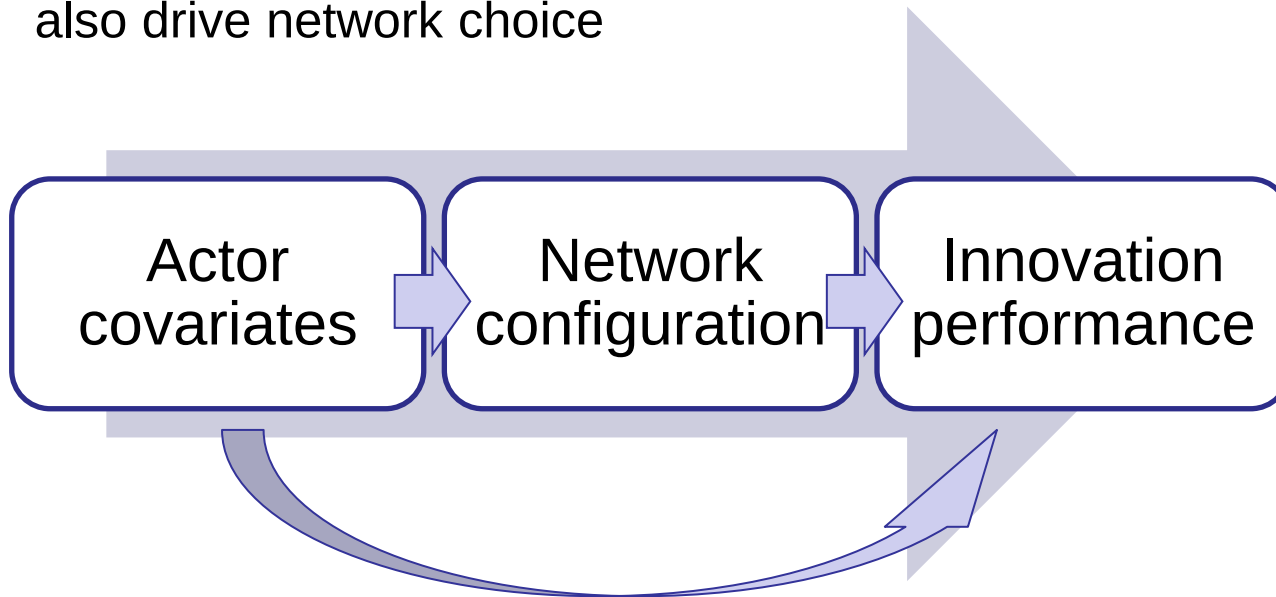
- Actor-level
- Dyadic level
 - Proximity
 - Homophily and similarity
- Network level (multi-network study)
 - The role of network content / setting

Endogeneity

COVARIATE DATA – ACTOR LEVEL

Dealing with endogeneity

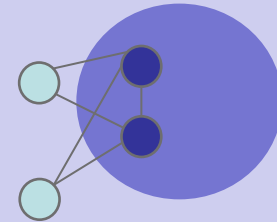
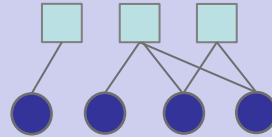
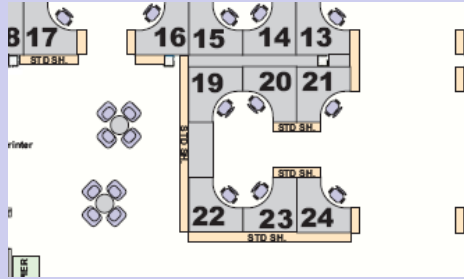
- Network choices are endogenous
 - The same factors leading to superior innovation performance may also drive network choice



- Recursive systems of equations? Longitudinal analysis?

Measuring proximity

COVARIATE DATA – DYADIC LEVEL



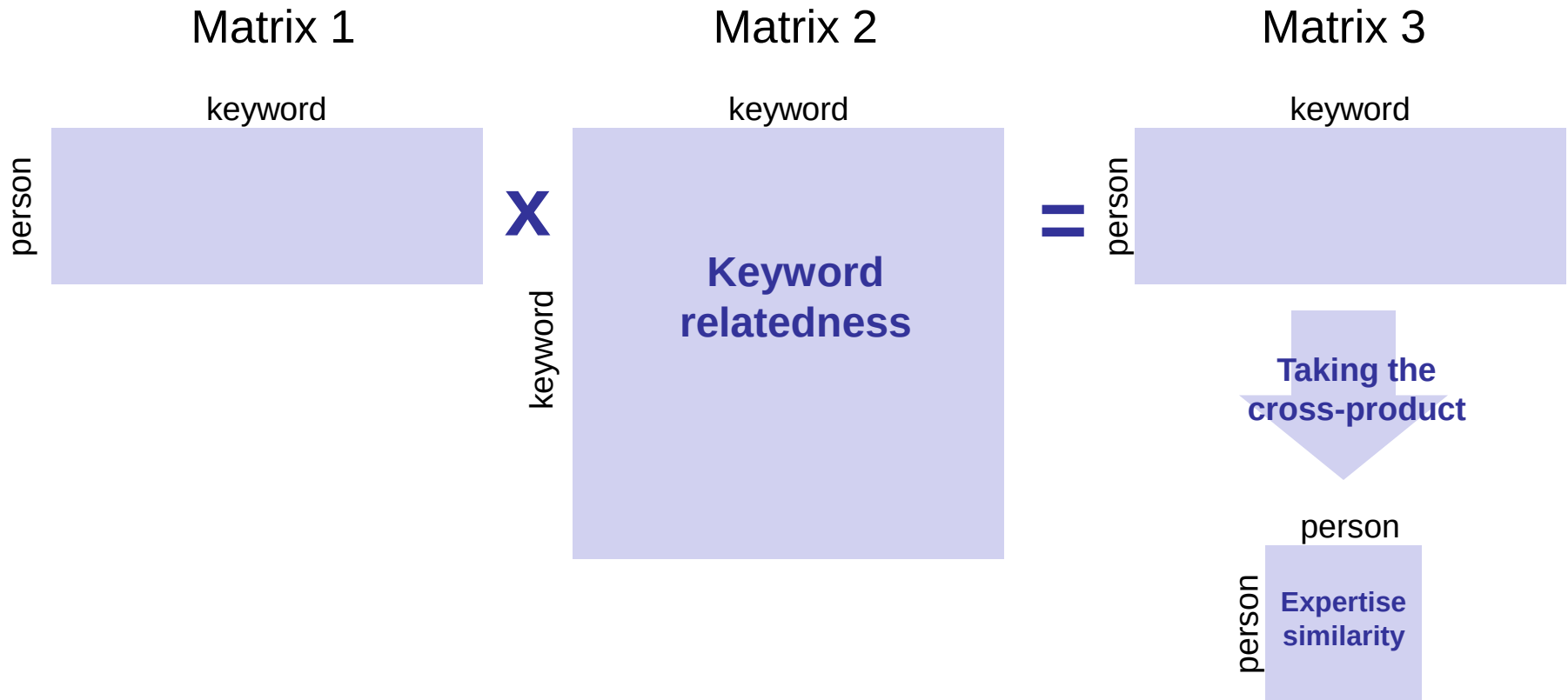
Physical
proximity

Organizational
proximity

Social
proximity

Measuring relatedness

COVARIATE DATA – DYADIC LEVEL



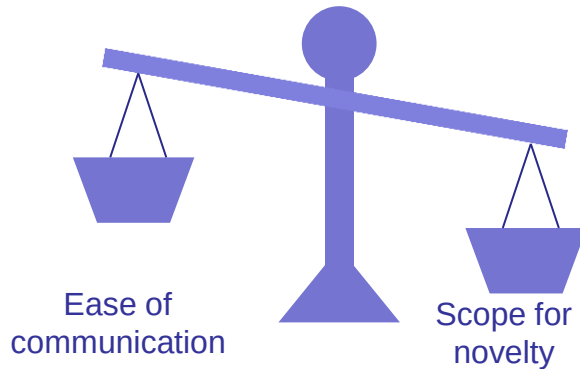
Network content

COVARIATE DATA – NETWORK LEVEL

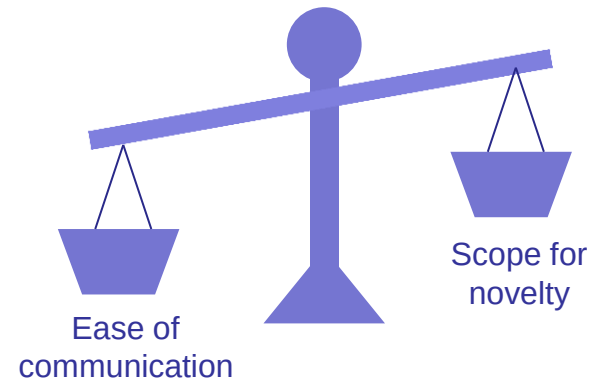
Information about the type of tie in the network.

Different types of networks may have different tie formation principles.

Idea generation



Implementation



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